MIL-F-51165B(EA)
25 May 1988
SUPERSEDING
MIL-F-51165A(EA)
10 December 1979

MILITARY SPECIFICATION

FILTER, PARTICULATE, 150 CFM, M9A1

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a particulate filter designed to remove particulate matter from atmospheric air.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-51079 - Filter Medium, Fire-Resistant, High-Efficiency.

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 123 - Marking for Shipment (Civil Agencies).

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance Test Methods.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-19-1821 - Filter, Particulate, 150 CFM, M9A1.
76-2-641-6 - Penetrometer, Filter Testing, DOP, Q107.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-1821 - Filter, Particulate, 150 CFM, M9A1.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-1821 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the filter shall conform to the specifications and drawings listed on Drawing 5-19-1821 and subsidiary drawings.

3.2 Manufacture and assembly. The filter shall be manufactured and assembled in accordance with Drawing 5-19-1821 (see 6.5).

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3.3 Resistance to airflow. The resistance to airflow through the filter shall not exceed 0.249 kilopascal (1.0 inch of water) at an airflow of 4.25 ± 0.15 cubic meters per minute (150 cubic feet per minute) when tested in accordance with 4.4.4.1.

3.4 DOP smoke penetration. The DOP smoke penetration of the filter shall not exceed 0.03 percent when tested as specified in 4.4.4.1.

3.5 Resistance to rough handling. The filter shall comply with the resistance to airflow and DOP smoke penetration requirements of 3.3 and 3.4 after rough handling for 15 minutes with pleats in a vertical position as specified in 4.4.4.2.

3.6 Workmanship. The filter shall be free from damage (bent, warped, or burred metal sections), and foreign matter (oil, grease, or dirt).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified hereon are classified as follows:

(a) First article inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection (see 6.2(c)).

4.3.1 Sample. The contractor shall submit a first article sample, packaged in accordance with SPI P5-19-1821, as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 10 packaged filters.

4.3.2 Inspections. As determined by the Government, the first article sample may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any filter fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any filter to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the particulate filters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of filter medium shall be represented in any one lot of particulate filters.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. The unpackaged filter shall be sampled in accordance with MIL-STD-105 for 4.4.3.4(a). For major defect 104 (resistance to rough handling), level S-4 shall be used. For the repair part acquisition, the packaged filter shall be sampled in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. For the repair part acquisition, the filter, unit packed in accordance with SPI P5-19-1821, shall be sampled in accordance with MIL-STD-105 for heat seal seam strength test.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample particulate filters shall be examined in accordance with 4.4.3.4(a) of the classification of defects and MIL-STD-105. For the repair part acquisition, the packaging inspection (4.4.3.4(b)) shall be conducted prior to the examination of particulate filters. Each unit packed filter shall be tested for unit pack bag leakage in accordance with 4.4.4.3. Any packaged filter failing to meet the unit pack bag leakage of SPI P5-19-1821 shall have the filter removed immediately from the lot, redried, repackaged and retested.

4.4.3.2 For destructive test. For the repair part acquisition, the sample of packaged filters shall be tested for heat seal seam strength (4.4.4.4), using MIL-STD-105 and an AQL of 1.0 percent.

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4.4.3.3 Critical defects. Each particulate filter in the lot shall be inspected for the critical defects listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Filter, particulate, 150 CFM, M9A1, assembly (Drawing 5-19-1821).

Categories	Defects	Acceptance standards
Critical:		
1	Resistance to airflow	4.4.4.1
2	DOP smoke penetration	4.4.4.1
Major: AQL 1.0 percent defective		
101	Damage or foreign matter (see 3.6)	
102	Marking incorrect or missing	
103	Component missing or incorrectly assembled	
AQL 1.5 percent defective		
104	Resistance to rough handling	4.4.4.2
Minor: AQL 2.5 percent defective		
201	Marking illegible	

(b) Packaging inspection (SPI P5-19-1821) (see section 6).

Categories	Defects	Acceptance standards
Critical:		
1	Unit pack container (bag) leakage	4.4.4.3
Major: AQL 4.0 percent defective		
101	Item not completely cleaned/dried	
102	Cushioning/blocking pads missing or not as specified	
103	Barrier missing or not as specified	
104	Unit container and closure damaged or not as specified	
105	Marking missing, incorrect or illegible	

4.4.4 Tests.

4.4.4.1 DOP smoke penetration and resistance to airflow. Filters shall be tested for DOP smoke penetration and airflow resistance with the Q107 penetrometer (76-2-641-6).

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4.4.4.2 Resistance to rough handling. The unpackaged sample filters, with pleats in a vertical position, shall be rough handled in accordance with Test Method T105.10 of MIL-STD-282 and then tested for DOP smoke penetration and resistance to airflow in accordance with 4.4.4.1.

4.4.4.3 Unit pack bag leakage. The leakage of the unit pack bag containing the filter shall be determined by subjecting the sample to the hot water technique, method 5009.2 of FED. TEST METHOD STD. NO. 101.

4.4.4.4 Heat seal seam. The heat seal seam strength of the unit pack bag shall be determined by subjecting the sample to FED. TEST METHOD STD. NO. 101, method 2024, except that the static load shall be 45 (+5, -0) ounces, and a 5 percent reduction shall be permitted when room temperature in test area exceeds 90°F.

5. PACKAGING

5.1 Interplant shipment (see 6.2 and 6.4). The filter shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in accordance with Fed. Std. No. 123, and shipping containers shall include the contractor's name, address, contract number, and date of manufacture.

5.2 Repair parts (see 6.2). When this item is being procured for storage and issue as a repair part, unit packing, packing, and marking shall be as specified in SPI P5-19-1821.

6. NOTES

6.1 Intended use. The particulate filter covered by this specification is intended for use in the ABCA-M6A1 filter unit.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of packaging (see 5.1 and 5.2).
(c) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(d) Shelf life code.

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6.3 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 296. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the prepared activity will take appropriate reconciliation action through international standardization channels including departmental standardization officer, if required.

6.4 Interplant shipment. Packaging and packing for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

6.5 Manufacturing suggestions. Filters manufactured by the following method have been satisfactory.

Pleat enough filter material (MIL-F-51079) on an appropriate pleating machine to give the desired compactness when fitted into the filter frame. Determine the correct number of pleats prior to the start of the operation by trial and error methods. The predetermined number of pleats should be strictly adhered to during operation. It is suggested that fiberboard separators be temporarily placed between pleats and that the pleated filter material be placed in a jig. The jig should hold the pleated filter material in the same state of compactness as will be required in the filter frame and allow access to both ends and one side of the pleated element. Withdraw the temporary fiberboard separators and replace with aluminum separators. To replace separators on the opposite side, turn the filter element over in the jig. When all aluminum separators are in place, trim the ends of the pleated filter to the desired width with a high speed band saw.

Coat the inside of the filter frame with sealing adhesive and insert the pleated element into the frame. On both sides of the filter, seal the entire seam between the filter element and the frame with sealing adhesive.

6.6 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word listing).

Airflow resistance
DOP smoke penetration
Filter, particulate
Media, filter
Penetrometer filter testing

Custodian:

Army - EA

Preparing activity:

Army - EA

Project 4240-A945

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51165B(EA)	FILTER, PARTICULATE, 150 CFM, M9A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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